

Interactive Peer Evaluation for Calibrated Trust in AI Advisers

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Related YIP white papers: Computational Cognition and Machine Intelligence (Dr. Ben Robinson);
Science of Information, Computation, Learning, and Fusion (Dr. Doug Riecken).

1. Research Problem and Central Idea

When several AI systems endorse the same answer, a user may reasonably expect more support than one system can provide. Yet those systems may repeat the same mistake. Consider one AI adviser that reports a shipment has arrived and three evaluators that endorse its answer. All cite summaries derived from one outdated notice; a revised log records a delay. Three endorsements therefore provide less corroboration than they appear to. Asking which dated source supports each judgment could expose the problem. Throughout, the *adviser* proposes an answer, the *panel* evaluates it, and the *evaluation record* reports endorsements, checked evidence, and unresolved concerns to the person deciding whether to rely on the advice.

This project asks: *How do people weigh agreement among AI evaluators, discount shared sources, and adjust reliance after errors?* Our central hypothesis is that disclosing shared sources and diagnostic evidence reduces harmful adoption of AI advice relative to a reliability estimate alone, without a practically meaningful loss of beneficial adoption. Aims 1 and 2 will produce and validate this evidence through dependence-aware peer evaluation and targeted follow-up questions; Aim 3 will test competing accounts of human reliance. The project addresses the Trust and Influence portfolio’s interests in drivers of trust, dynamic trust assessment, and human-machine joint performance. Its behavioral focus is the establishment, maintenance, and repair of trust in an AI adviser.

We distinguish three concepts. **Reliability** is the measured quality of advice on a specified task. **Trust** is a person’s expectation that the adviser will help, measured through reported beliefs. **Reliance** is the decision to use its advice. Trust can guide reliance without guaranteeing that reliance is appropriate [15]. Following work on appropriate reliance [19], we will measure beneficial acceptance of correct advice and resistance to incorrect advice separately. Higher reported trust or more frequent agreement with AI will not by itself establish improvement.

Foundation and scientific gap. The PI’s PRD method aggregates peer models’ pairwise judgments through iterative ranking and studies discussion between evaluators [17]. IQA-EVAL generates and assesses multi-turn question-answering interactions, with persona-based evaluation agents validated against human ratings [16]. These supply complementary starting points: combining imperfect judgments and testing an adviser through interaction. Neither result establishes that peer scores are probabilities of correctness or that simulated users reproduce human trust.

Related research documents correlated model errors and the limits of evaluation panels [12, 13]. Human-AI studies show that explanations and interaction design affect reliance, with persistent errors even in cooperative question answering [2, 11]. Behavioral research finds an illusion of consensus even when people can identify shared sources [26]; other studies show sensitivity to shared evidence and clearer source-independence cues [4, 24]. With labeled source relations and an always-correct simulated adviser, Ueno et al. found greater reliance on independent than shared-source support [22]. Study 1 tests whether

this distinction holds when evaluators can be wrong, measuring harmful and beneficial adoption separately. We connect the evidence a panel supplies with the corroboration people perceive, then test reliance after diagnostic interaction and subsequent failures.

2. Research Aims and Technical Approach

Aim 1: Estimate what peer agreement actually tells us

We will extend PRD from relative model ranking to assessing particular advice, building on work with panels of LLM judges [23]. Evaluators will first judge candidate answers independently, with identities hidden and answer order randomized, and record supporting passages and unresolved objections. We will retain these initial judgments before any discussion so that later agreement can be distinguished from initial corroboration.

Classical models estimate observer error rates [5], jointly infer expertise and item difficulty [25], and discount copied sources in truth discovery [9]. Building on these foundations, we will fit a regularized model of reviewer skill, task difficulty, and shared source effects on independently adjudicated development cases. A starting method will adjust peer weights using these estimates and group endorsements that rely on the same source. This separates an evaluator’s demonstrated judging ability from its rank as an answer generator. Source overlap is observable evidence of dependence; different model names or prompts will not be treated as proof of independence. Comparisons will include observer-error and expertise/difficulty models without source terms, copied-source weighting, unmodified PRD, majority vote, the development-selected best judge, and supervised aggregation given the same labels. The proposed advance links provenance-based assessment of LLM evaluator dependence to its effects on human reliance.

The output will combine an estimated probability that the advice is correct and supported by the supplied documents with its evidence and unresolved concerns. Calibration will use data separate from both method development and final testing. In the shipment example, three references to one notice will be distinguished from support drawn from separately checked records. Controlled source duplication and measured model errors will test when this distinction improves assessment. Without suitable reference judgments, agreement can support a relative comparison but cannot establish absolute correctness. Tests on new topics will assess where calibration fails and additional human checking is needed. The research question is how much independent checking is needed before a panel’s agreement becomes a useful reliability signal.

Aim 2: Ask questions that reveal weaknesses in the advice

We will extend IQA-EVAL’s interaction generator to select short diagnostic exchanges. Prior LM cross-examination detects factual errors through inconsistencies [3]; our extension targets provenance and recency, separating detection, correction, and harm before testing human reliance. Candidate questions

will request a precise supporting passage, challenge an apparent contradiction, or ask whether a newer record changes the answer. In the shipment example, the useful question concerns the date and source of the arrival claim. A request to explain the same answer more fluently may add no evidence.

The system will select up to three follow-up questions based on how often each type exposes errors on similar development cases and how much interaction it requires. Randomized trials of a finite set of question types will estimate which checks work for each failure type; fixed scripts and random selection provide baselines. Reference labels will remain hidden during evaluation. We will score the original advice and any revision separately: detecting an error, correcting it, and persuading another evaluator are different outcomes.

We hypothesize that questions seeking new, relevant evidence will expose shared mistakes more effectively than requests for additional agreement. Comparisons will include independent peer judgments, ordinary peer discussion, extra samples from one model, and targeted questions with the same available documents and matched computational budgets. The evaluation record will state what was checked, which answer changed, and what remains unresolved. We will also measure damage from inappropriate challenges, including correct advice changed to an incorrect answer. Unjustified agreement and answer-flipping after challenges are documented risks [14, 20]. IQA-EVAL agents will help develop interactions; human experiments in Aim 3 will test their consequences for trust and reliance.

Aim 3: Explain how evaluation evidence changes human reliance

We will conduct two preregistered online studies using short document-based decisions with independently established answers. Following the judge-advisor paradigm [1], participants will first give their own answer and confidence, then review AI advice and an evaluation record before deciding again. This permits measurement of reliance when the person and adviser initially disagree, rather than inferring reliance from a correct final answer alone.

Study 1: AI consensus as a social cue. We will vary shared versus separately sourced support and randomly assign participants to three displays: an endorsement count, a calibrated reliability estimate, and the same estimate accompanied by checked sources and unresolved objections. Display varies between participants; sourcing, advice correctness, and task type vary within participants. Advice content, correctness balance, panel size, and allocated reading time will be matched across display conditions. A targeted contrast will hold the advice, endorsements, and numerical estimate fixed while revealing that endorsements reuse one source. Comprehension checks will test whether participants recognize the dependence. This isolates the behavioral effect of disclosure from improvements in the advice itself. Direct record-lookup and multi-report integration tasks will test whether the effect depends on how readily claims can be checked.

The primary contrast tests whether adding records to the estimate reduces harmful adoption more for shared than separate sources. Success requires lower harmful adoption on shared-source cases and

beneficial adoption non-inferior to the estimate-only display within a margin fixed before confirmatory collection. Competing predictions are that the illusion persists despite comprehension [26], or that richer records make wrong advice more persuasive, as observed with erroneous LLM explanations [21]. The balanced-correctness design tests this risk separately from beneficial adoption.

Study 2: Trust repair after errors. Seeing an algorithm err can induce avoidance [8], motivating established work on trust repair in human-autonomy interaction [6, 18]. Across repeated decisions, participants will receive feedback about outcomes. After a disclosed failure, we will randomly assign a corrected answer alone or the same correction plus the diagnostic evidence that led to it. Subsequent cases will include both recurrence of the failure type and unrelated reliable advice. We will test whether the evidence supports selective adjustment rather than indiscriminate acceptance or rejection of the adviser. Explanations of why an aid errs can increase reliance even when trust is unwarranted [10]; diagnostic evidence might therefore restore reliance globally, including on recurring failures. The recurrence-versus-unrelated contrast distinguishes these predictions. An initial planning range is 200–300 participants per study; pilot-based power simulations for the primary contrasts and Study 1’s non-inferiority criterion will determine final sample sizes.

Building on longitudinal trust-calibration theory [7], a sequential behavioral model will predict reliance from prior outcomes, self-confidence, estimated advice reliability, and perceived independent corroboration. Competing accounts will treat endorsements as separate votes or discount their common sources. Their predictions will be compared on held-out participants and tasks against models based only on agreement counts or past accuracy. The key test is whether dependence and diagnostic evidence explain reliance beyond these simpler accounts. Brief trust ratings and forecasts of adviser accuracy will measure beliefs separately from choices. Randomized displays support causal tests of interventions; associations involving reported beliefs will not alone establish a causal mechanism.

3. Evaluation, Feasibility, and Expected Outcomes

Computational evaluation will start with bounded evidence dossiers for scheduling and report reconciliation, including outdated records, duplicated sources, and unresolved conflicts. Human reviewers will establish answers and evidence support without seeing evaluator identities or conditions. Ambiguous cases will be retained as unresolved rather than assigned artificial ground truth. Held-out documents, topics, and evaluator combinations will test transfer. We will report calibration error, error-detection precision and recall, harmful revisions, abstention, and total cost, including all model calls and checks. Reliability comparisons will match the fraction of cases receiving an assessment. Separate label-budget curves will show the cost of external calibration.

Human evaluation will report beneficial adoption when the initial human answer is wrong and AI advice is correct, and harmful adoption when the human is right and AI advice is wrong. Cases where

both are wrong will be analyzed separately. Final accuracy, time, workload, and reported trust will complement these measures. Analyses will account for repeated observations by participant and task, report uncertainty intervals, and preregister primary contrasts. We will distinguish controlled diagnostic cases from tests at the observed error prevalence. Participant studies will undergo institutional review and use informed consent. Failure to improve reliance despite improved automated assessment is an informative result about the limits of communicating evaluation evidence.

Team and work plan. PI Xinya Du will lead the work with graduate researchers at UT Dallas. The proposed three-year budget of \$450,000, including indirect costs, supports personnel, computation, annotation, participant compensation, and dissemination. IQA-EVAL's validation against human ratings provides experience connecting automated evaluations with human judgments. The PI will seek consultation on behavioral protocols and power analysis before confirmatory data collection. **Year 1** will establish the reference cases, dependence measurements, and calibrated peer assessments. **Year 2** will develop diagnostic interactions and conduct Study 1. **Year 3** will conduct Study 2 and test transfer of the computational and behavioral models. PeerEval, the PI's prototype peer-evaluation leaderboard, provides a starting point; interfaces for the human studies will be developed.

Potential Air Force and Space Force impact. Applications could include AI-assisted reconciliation of logistics updates, maintenance records, and mission-status reports. Multiple evaluators may appear to corroborate an interpretation while relying on the same upstream report. This research would identify when to display an endorsement count, disclose shared sources, or request an additional check. It would also provide measures for testing whether those interventions reduce harmful reliance while preserving useful advice after a visible failure. These are prospective applications; operational transfer would require evaluation with the relevant users and records.

Expected outcomes are testable accounts of how corroboration affects reliance, algorithms for interactive peer evaluation, and reusable tasks and study protocols. Together, these would establish when automated evaluation supports human judgment and when additional independent evidence is necessary.

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